Foundations of Data Science, Artificial Intelligence, and Emerging Paradigms

Comprehensive Beginner Handout for BSIT Data Science

Data Foundations • ML and DL • Generative AI • Agentic AI • Responsible Use

Contents

1 The Big Picture: Data Science, AI, and Emerging Paradigms

Data Science

Data Science is an interdisciplinary field that combines **statistics, computing, data engineering, machine learning, domain knowledge, and communication** to extract useful insights, build predictive systems, and support better decisions from data.

Artificial Intelligence

Artificial Intelligence (AI) is the broad field of computer science focused on building systems that can perform tasks that usually require human intelligence, such as reasoning, learning, perception, language understanding, and decision-making.

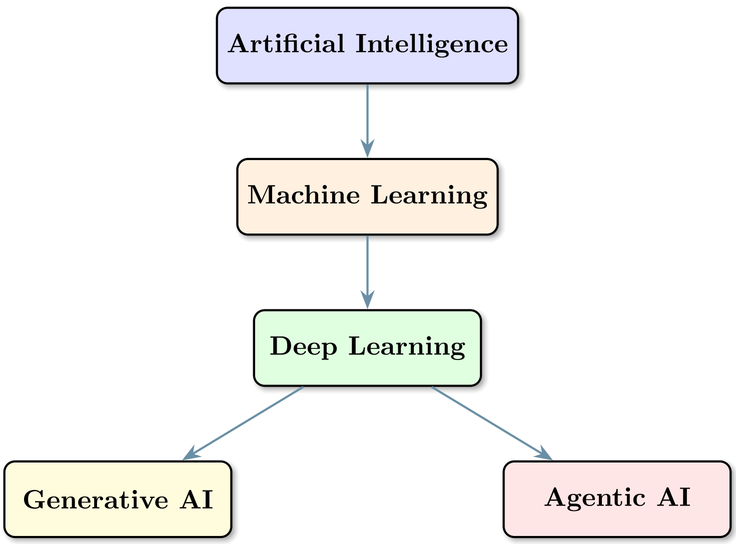
How These Fields Relate

Data Science focuses on turning raw data into understanding and action. AI focuses on building intelligent behavior. In practice, the two fields overlap strongly because Data Science often uses AI techniques, while AI systems depend on strong data foundations.

1.1 Common Terms in This Handout

Field or Term	Main Focus
Data Science	End-to-end process of collecting, cleaning, analyzing, modeling, and using data
Data Mining	Discovering patterns and relationships from data
Machine Learning	Learning patterns from data to make predictions or decisions
Deep Learning	Using multi-layer neural networks for complex learning tasks
Generative AI	Creating new content such as text, images, code, or audio
Agentic AI	Building goal-directed systems that can reason, plan, and act

1.2 AI as the Umbrella Field



- **Machine Learning** is a major subfield of AI.
- **Deep Learning** is a specialized branch of machine learning.
- **Generative AI** and **Agentic AI** are important emerging paradigms built on modern AI and deep learning methods.

2 Foundations of Data Science

2.1 Why Data Science Matters

- It helps organizations and researchers turn raw data into evidence.
- It supports descriptive, predictive, and prescriptive decision-making.
- It combines technical methods with domain understanding and communication.
- It provides the full pipeline from data collection to deployment and monitoring.

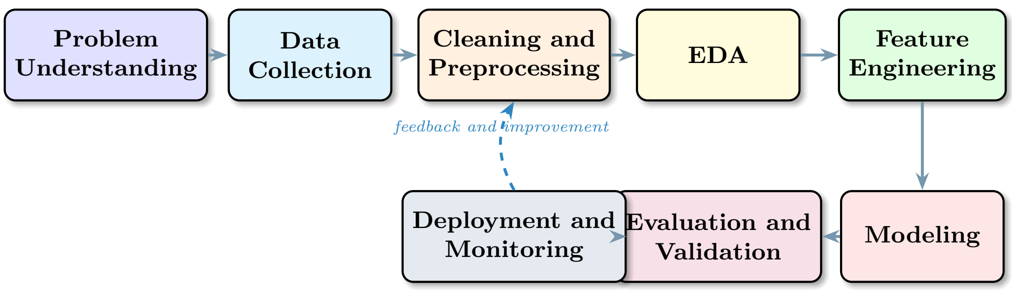
2.2 Data Mining vs. Data Science

Aspect	Data Mining	Data Science
Main goal	Discover hidden patterns	Deliver end-to-end value from data
Main focus	Pattern discovery and knowledge extraction	Full lifecycle including modeling, deployment, and decision support
Typical scope	Analysis-centered	Analysis plus engineering, automation, and communication
Relationship	Important component	Broader field that includes data mining

Key Point

Data mining is not separate from Data Science. It is one important part of the broader Data Science lifecycle.

2.3 The Extended Data Science Lifecycle



1. **Problem or business understanding:** Define the question and the goal.
2. **Data collection:** Gather raw data from relevant sources.

3. **Data cleaning and preprocessing:** Handle missing values, duplicates, errors, and inconsistent formats.
4. **Exploratory Data Analysis (EDA):** Summarize, visualize, and understand the data.
5. **Feature engineering:** Transform data into better inputs for modeling.
6. **Modeling:** Apply machine learning or deep learning methods.
7. **Evaluation and validation:** Test whether the solution works reliably.
8. **Deployment and monitoring:** Put the solution into use and track its performance over time.

Why the Lifecycle Matters

The lifecycle shows that Data Science is not just about building a model. It is about creating a complete system that solves a real problem and remains useful after deployment.

3 Core Data Foundations

3.1 Dataset, Subject, and Observation

Dataset

A dataset is a structured, semi-structured, or unstructured collection of observations that represents real-world entities, events, or processes.

- **Subject or entity of analysis:** What the dataset is about.
- **Observation or instance:** One row, record, case, or sample in the dataset.

Examples:

- Students in an educational analytics dataset
- Patients in a healthcare dataset
- Customers in a business dataset

Important Modeling Principle

Correctly identifying the subject and observation unit is essential. If the unit of analysis is wrong, the model and conclusions may also be wrong.

3.2 Structure of Data

Data Form	Main Characteristic	Examples
Structured data	Fixed rows and columns	CSV files, Excel sheets, relational databases
Semi-structured data	Flexible schema, tagged or nested format	JSON, XML, logs, API responses

Data Form	Main Characteristic	Examples
Unstructured data	No fixed schema	Text, images, audio, video

Modern AI and Data

Classical machine learning often works well on structured data, while modern AI and deep learning are especially important for semi-structured and unstructured data such as text, images, and audio.

3.3 Features and Target Variables

Features

Features are the measurable characteristics or attributes that describe an entity. They are usually the input variables used by a model.

Target Variable or Label

The target variable is the outcome that we want to predict or explain.

Example for a student dataset:

- **Features:** Attendance rate, prior GPA, LMS activity time
- **Target:** Final GPA, pass or fail, dropout risk

Feature Engineering

Feature engineering is the process of selecting, transforming, or creating better features so that a model can learn more effectively.

3.4 Measurement Scales and Data Types

Type	Meaning	Why It Matters
Nominal	Unordered categories	Requires encoding because categories have no numeric order
Ordinal	Ordered categories	Order matters, but numerical distance may not
Discrete numerical	Countable values	Common in counting tasks and event-based data
Continuous numerical	Real-valued measurements	Important for regression, scaling, and distance-based methods

- Understanding data types helps with algorithm selection.
- It also affects encoding, distance computation, and data preprocessing.

3.5 Labeled, Unlabeled, and Semi-Supervised Data

- **Labeled data:** Features are paired with a known output. Used in supervised learning.
- **Unlabeled data:** No target variable is available. Used in unsupervised learning.
- **Semi-supervised learning:** Combines a small amount of labeled data with a larger amount of unlabeled data.

3.6 Primary and Secondary Data Sources

- **Primary data:** Collected directly for the current purpose. Usually offers more control, but takes more time and cost.
- **Secondary data:** Already exists from previous studies, public repositories, organizations, or systems.

3.7 Other Essential Beginner Concepts

- **Missing values:** Data fields that are empty or unknown.
- **Duplicates:** Repeated records that may distort analysis.
- **Outliers:** Extreme values that may be unusual or informative.
- **Train, validation, and test sets:** Separate data portions used to train, tune, and evaluate a model fairly.

4 Levels of Analytics in Data Science

Analytics Level	Guiding Question	Typical Methods
Descriptive / exploratory	What happened?	Summary statistics, tables, charts, EDA
Diagnostic	Why did it happen?	Root-cause analysis, comparisons, drill-down analysis
Predictive	What will happen?	Classification, regression, forecasting, ML and DL
Prescriptive	What should we do?	Optimization, recommendation, simulation, decision support

Analytics Maturity

As we move from descriptive to prescriptive analytics, the focus shifts from understanding the past to guiding future decisions and actions.

5 Machine Learning, Deep Learning, and Data Mining

5.1 Machine Learning

Machine Learning

Machine Learning (ML) enables systems to learn patterns from data and make predictions or decisions without being explicitly programmed with fixed rules for every situation.

Common machine learning tasks:

- **Classification:** Predicting categories such as pass or fail, spam or not spam.
- **Regression:** Predicting numerical values such as income, GPA, or price.
- **Clustering:** Grouping similar items together when labels are not available.

5.2 Deep Learning

Deep Learning

Deep Learning is a specialized branch of machine learning that uses neural networks with many layers to learn complex and hierarchical patterns from large datasets.

Deep learning powers:

- Large Language Models (LLMs)
- Image generation systems
- Speech recognition systems
- Modern generative models

Simple Comparison

Classical machine learning often depends more heavily on manual feature engineering, while deep learning can automatically learn complex representations from raw or minimally processed data.

5.3 How Data Mining Fits Here

Data Mining

Data Mining focuses on discovering meaningful patterns, trends, correlations, or structures from large datasets using statistical and machine learning techniques.

- Its main emphasis is **pattern discovery**, **knowledge extraction**, and **interpretation**.
- It is highly relevant during EDA and pattern analysis within the Data Science workflow.

6 Generative AI

Generative AI

Generative AI refers to models that learn the underlying patterns or probability distribution of data and then generate new outputs that resemble the training data, such as text, images, code, audio, or video.

Generative vs Predictive

Predictive systems estimate labels or values. Generative systems create new content.

6.1 Core Characteristics of Generative AI

- Learns data distributions and patterns
- Produces novel outputs rather than only class labels
- Often behaves probabilistically, meaning outputs can vary
- Usually depends on very large-scale datasets and deep learning models

6.2 Common Generative Model Families

Model Family	Typical Output	Key Idea
Autoregressive models	Text, code, sequences	Generate outputs step by step, often token by token
Variational Autoencoders (VAEs)	Synthetic samples, compressed representations	Learn a latent representation and generate from it
Generative Adversarial Networks (GANs)	Synthetic images and samples	A generator and discriminator learn through competition
Diffusion models	Images, audio, video	Start from noise and refine it into meaningful output

6.3 Important Beginner Terms for Generative AI

- **Prompt:** The instruction given to the model.
- **Token:** A small unit of text processed by the model.
- **Context window:** The amount of information the model can consider at one time.
- **Inference:** The stage when the trained model generates an answer for a new input.

6.4 GitHub Copilot as Generative AI in Practice

GitHub Copilot

GitHub Copilot is an AI-powered code generation assistant integrated into development environments such as VS Code. It uses deep learning and large language models to help users write, explain, and improve code.

For Data Science work, GitHub Copilot can:

- Suggest data preprocessing code
- Help write feature engineering steps
- Generate visualization scripts

- Assist with model implementation
- Explain functions and error messages
- Speed up experimentation and prototyping

6.5 Common Copilot Modes in Modern Editors

Mode	What It Does	Typical Student Use
Ask mode	Explains concepts, code, or errors	Learn syntax, understand models, ask questions
Edit mode	Applies requested code or text changes	Refactor notebooks, adjust scripts, rewrite blocks
Agent mode	Works through multi-step tasks using tools and repo context	Search files, update several places, validate changes

Important Reminder About Copilot

GitHub Copilot suggestions must always be reviewed, tested, and understood. Generated code can still contain errors, bias, or insecure practices.

7 Agentic AI

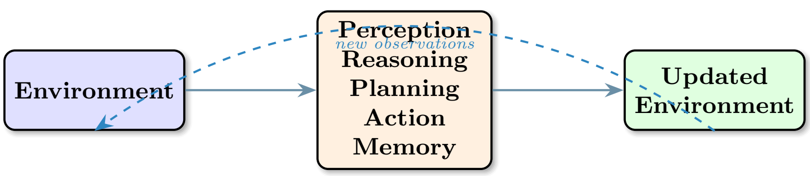
Agentic AI

Agentic AI refers to autonomous or semi-autonomous AI systems that can perceive inputs, reason about goals, plan actions, execute decisions, and adapt over time.

7.1 Core Components of an AI Agent

- **Perception:** Observing input, context, or environment
- **Reasoning:** Interpreting the situation and choosing what makes sense
- **Planning:** Sequencing steps toward a goal
- **Action:** Executing tasks through tools, APIs, or generated outputs
- **Memory:** Keeping useful context for future steps

7.2 Agentic Feedback Loop



7.3 Applications of Agentic AI

- Autonomous research assistants
- Intelligent tutoring systems
- Workflow automation agents
- Academic advising systems
- Decision-support systems

Relationship to Generative AI

Agentic AI often uses a generative model such as an LLM as its reasoning engine, but extends it with planning, memory, tools, and action loops.

8 Generative AI vs. Agentic AI

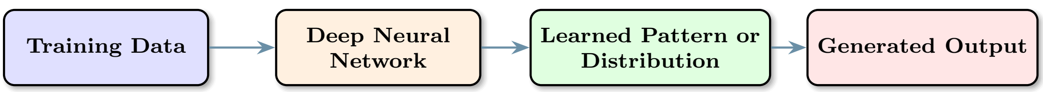
Aspect	Generative AI	Agentic AI
Primary focus	Content generation	Goal-driven action
Core capability	Generate text, images, code, or audio	Decide, plan, and act
Autonomy	Often limited	Higher and more persistent
Interaction style	Prompt-based	Environment-based or tool-based
Memory	Often short-term	Often persistent or state-aware
Typical output	Content	Actions, plans, decisions, and content

Integration Principle

Generative AI often acts as the cognitive engine inside an agentic system, while agentic AI adds planning logic, tools, memory, and goal management around that engine.

9 Architectural Perspectives

9.1 Generative Model Flow



9.2 Combined Agentic Architecture

Future Direction

Modern intelligent systems increasingly combine strong data foundations, predictive modeling, generative capabilities, and agentic behavior into practical, scalable, and domain-aware solutions.

10 Ethical and Practical Considerations

Issue	What It Means	Why It Matters
Bias and fairness	Outputs may reflect unfair patterns in data	Can harm people and reduce trust
Hallucination and reliability	Models may produce believable but false content	Can mislead decisions and reports
Privacy and security	Sensitive data may be exposed or mishandled	Important in education, health, and business contexts
Over-reliance on AI	Users may accept outputs without review	Human understanding and validation remain essential
Responsible deployment	Systems need limits, testing, and governance	Important for education, research, and real-world use

Educational Responsibility

In BSIT and Data Science courses, AI tools can support learning, but they should not replace understanding. Students are responsible for reviewing generated code, verifying answers, and explaining their own work.

Human-in-the-Loop

Human-in-the-loop means that people remain involved in checking, approving, or correcting AI decisions, especially when the results matter for grades, academic work, policy, security, or public use.

11 Integrated Summary

- Data Science provides the structured lifecycle for transforming raw data into useful insights and deployed systems.
- Data Mining focuses on discovering patterns and is one component within the broader Data Science workflow.
- Machine Learning and Deep Learning power predictive and intelligent systems.
- Generative AI creates new content such as text, images, and code, with GitHub Copilot as a practical coding example.
- Agentic AI builds goal-directed systems that perceive, reason, plan, act, and adapt.
- The future of intelligent systems lies in combining strong data foundations, robust models, generative capabilities, and agentic architectures in responsible ways.

12 Review Questions

1. What is the difference between Data Science and Data Mining?
2. Why is Artificial Intelligence described as an umbrella field?
3. What are the common tasks in machine learning?
4. How does deep learning differ from classical machine learning?
5. Why is it important to identify the subject and observation unit correctly?
6. What is the difference between structured, semi-structured, and unstructured data?
7. How do descriptive, predictive, and prescriptive analytics differ?
8. What makes Generative AI different from predictive AI?
9. How is GitHub Copilot an example of Generative AI in practice?
10. What are the core components of an AI agent?
11. How does Agentic AI differ from Generative AI?
12. Why are ethics, privacy, and human review important in Data Science and AI systems?